

## Practical - 3

### Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

#### F. Theory

1. How does the layout adapt for mobile, tablet, and desktop screens?

Answer: A layout adapts across mobile, tablet, and desktop screens using fluid grids, flexible images, and breakpoints (typically under 600px for mobile, 601px–1024px for tablets, and 1025px+ for desktops). Elements reflow, resize, or stack vertically to fit the available screen space.

(<https://developer.mozilla.org/en>

[US/docs/Learn\\_web\\_development/Core/CSS\\_layout/Responsive\\_Design](https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design))

2. Which layout technique is used and why?

Answer: Common web design layout techniques include **CSS Flexbox**, **CSS Grid**, and traditional **Float** or **Responsive** structures. Because your specific project or graphic was not provided, the correct technique depends entirely on the design goal.

([https://www.w3schools.com/html/html\\_layout.asp](https://www.w3schools.com/html/html_layout.asp))

3. Are typography, spacing, colors, and contrast readable?

Answer: Yes, typography, spacing, colors, and contrast directly determine whether text is readable. When properly balanced, they prevent eye strain and make content easy to scan. When poorly configured, text becomes difficult or impossible to read.

(<https://guidelines.india.gov.in/activity/the-impact-of-typography-on-web-design-a-comprehensive-guide-to-aesthetics-and-readability/>)

4. Is the framework used consistently without unnecessary inline CSS?

Answer: `style=""` attributes. While they look like inline styles in the markup, they compile into clean external stylesheets and maintain design consistency without unnecessary true inline CSS.

([https://www.reddit.com/r/Frontend/comments/rk3lr4/are\\_css\\_frameworks\\_really\\_necessary/](https://www.reddit.com/r/Frontend/comments/rk3lr4/are_css_frameworks_really_necessary/))

**G. Implementation/ Output/Result :****- CSS Selectors Used:**

\*, body, header, nav, main, section, article, footer, div,  
h1, h2, h3, h4, p, ul, li, a, img,  
form, label, input, textarea, select, button,  
table, tr, td, hr, small, i, marquee,  
.class, #id,  
:hover, :focus, :first-child, :last-child,  
input[type="text"], input[type="email"], input[type="password"],  
input[type="radio"], input[type="checkbox"],  
input[type="submit"], input[type="reset"],  
@media

**- CSS Properties Used:**

margin, padding, box-sizing,  
display, width, height, max-width,  
background, background-size, background-position, background-repeat,  
color,  
font-family, font-size, font-weight,  
text-align, text-decoration, line-height,  
border, border-left, border-top, border-radius,  
box-shadow,  
position,  
justify-content, align-items,

flex-direction, flex-wrap,

gap,

grid-template-columns,

list-style,

cursor,

transition,

transform,

outline,

resize,

object-fit,

overflow

## H. Evidence of Work (Screenshots)

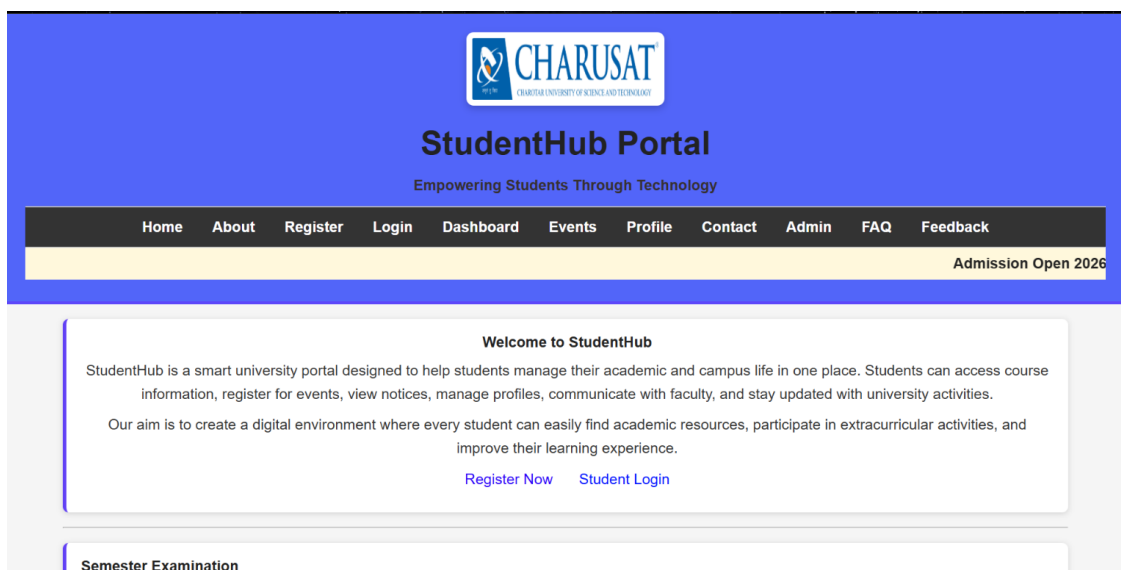


Figure 3.1(Home page)

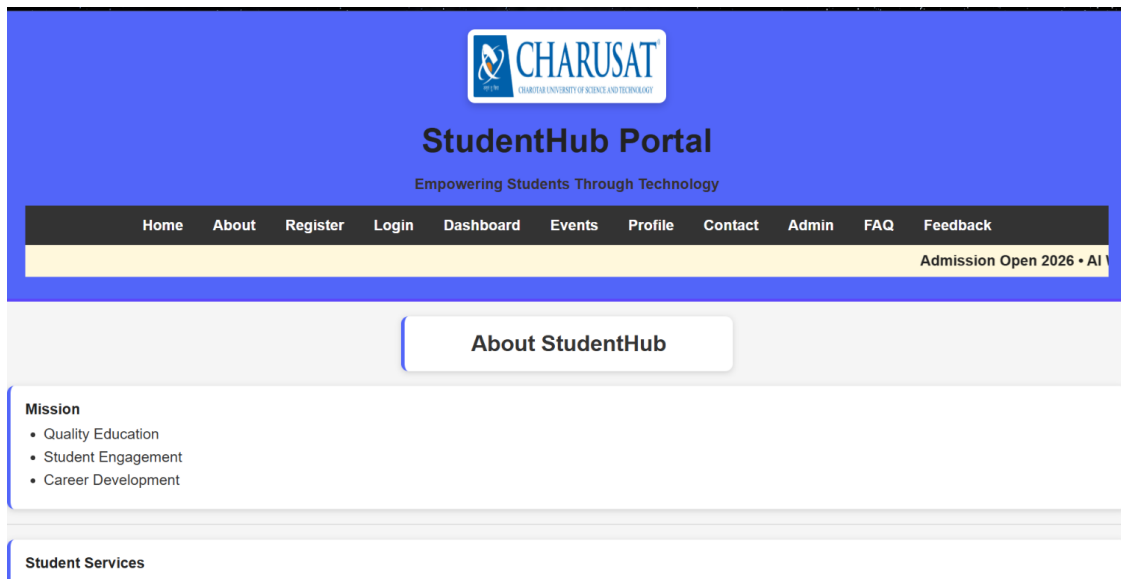


Figure 3.2(About page)

Figure 3.3(Registration page)

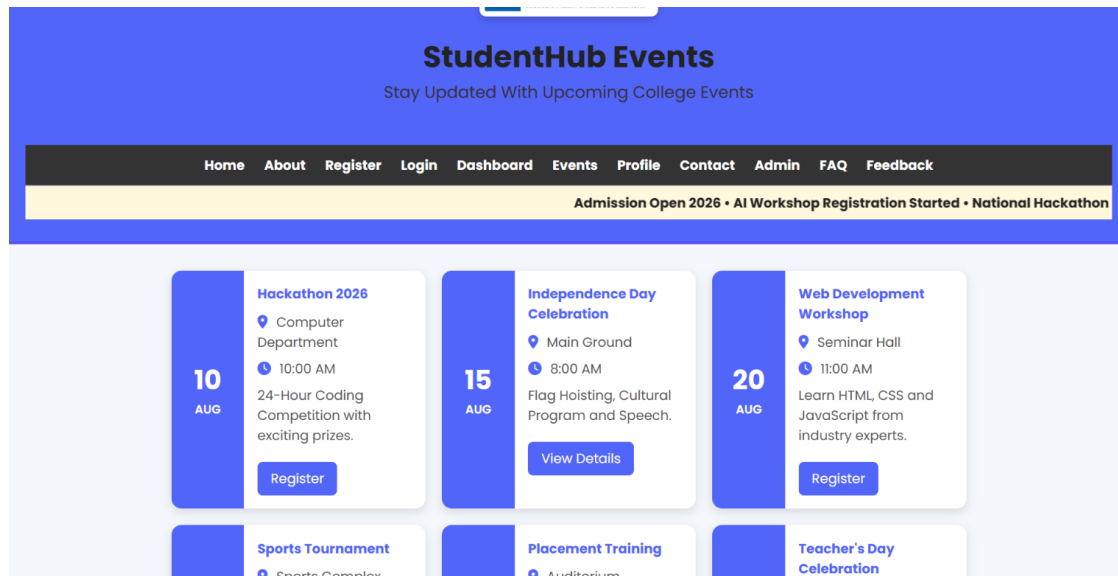


Figure 3.4(Event page)

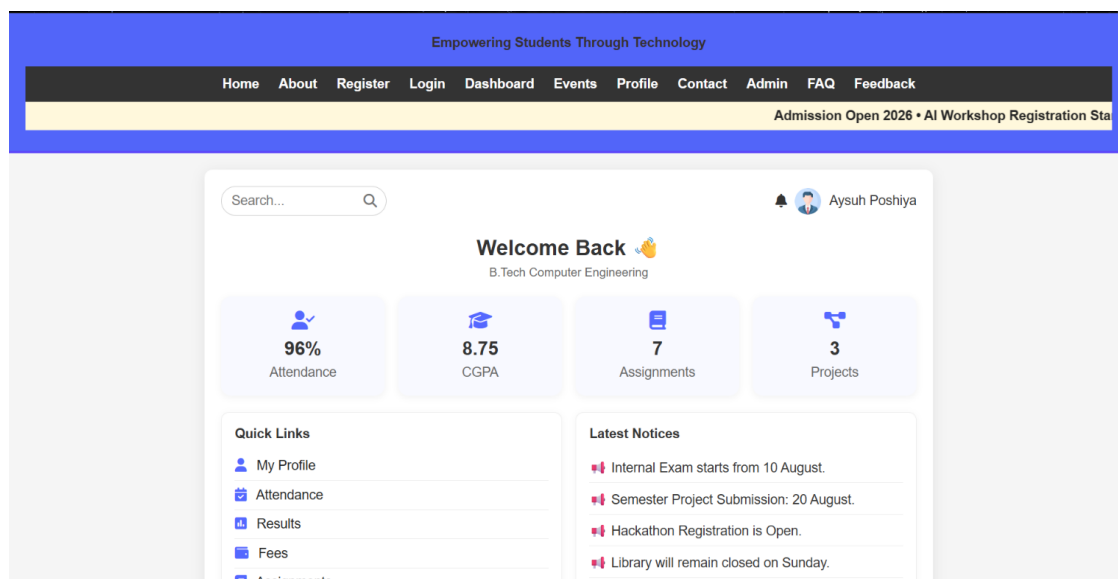
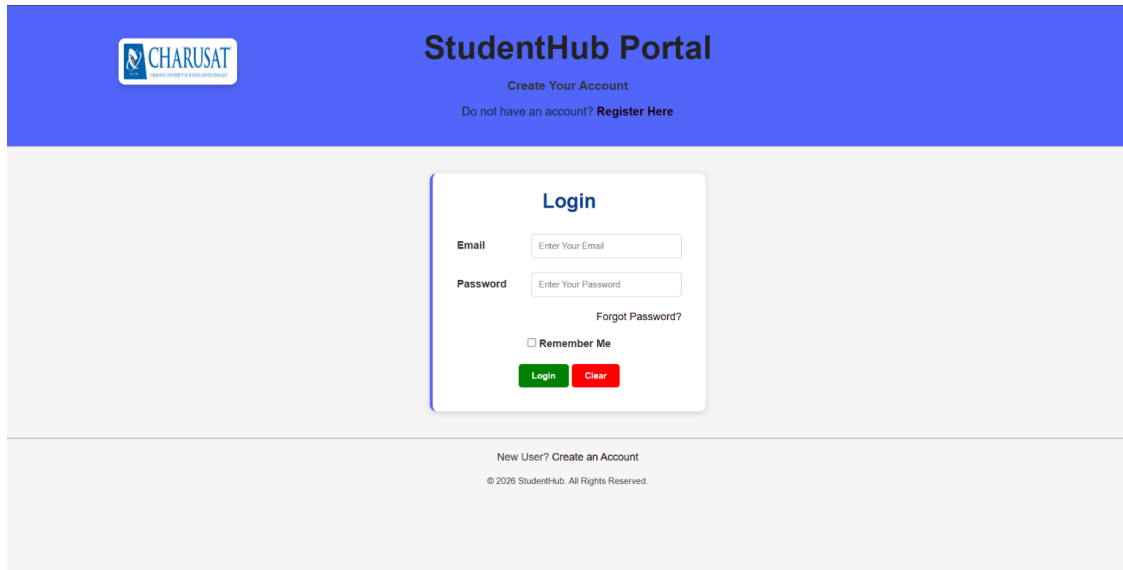
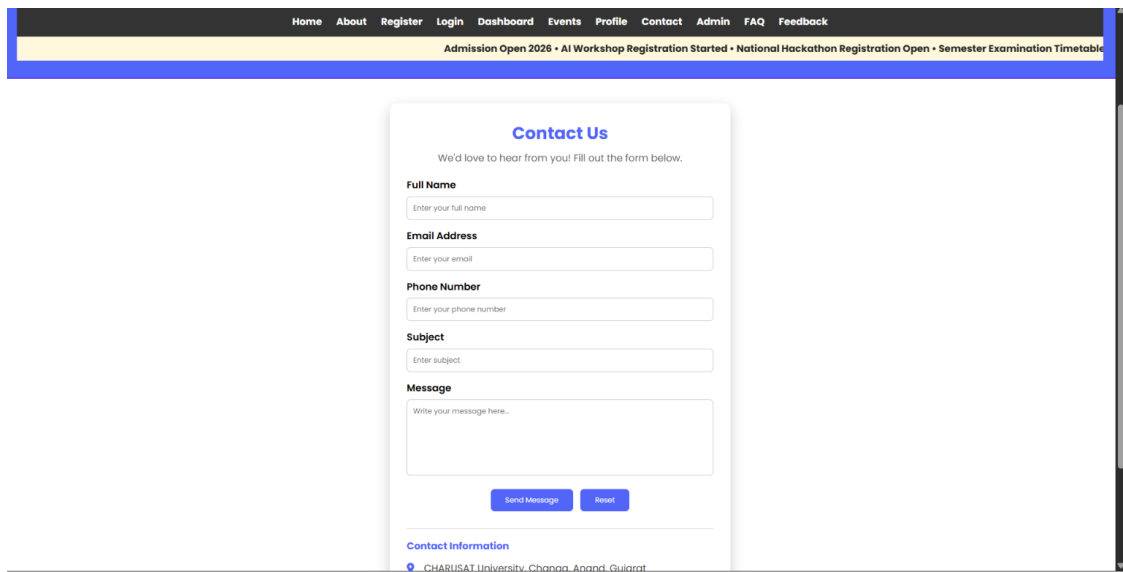


Figure 3.5(Dashboard page)



The image shows the login page of the StudentHub Portal. At the top, there is a blue header with the CHARUSAT logo on the left and the text "StudentHub Portal" in the center. Below the header, there are links to "Create Your Account" and "Do not have an account? Register Here". The main content area is light gray and contains a white login form. The form has a title "Login" and two input fields: "Email" and "Password". Below the password field is a link "Forgot Password?". There is a checkbox for "Remember Me" and two buttons: "Login" (green) and "Clear" (red). At the bottom of the page, there is a link "New User? Create an Account" and a copyright notice "© 2026 StudentHub. All Rights Reserved."

Figure 3.6( login page)



The image shows the contact page of the StudentHub Portal. At the top, there is a dark blue header with a navigation menu: "Home", "About", "Register", "Login", "Dashboard", "Events", "Profile", "Contact", "Admin", "FAQ", and "Feedback". Below the header is a yellow banner with the text "Admission Open 2026 • AI Workshop Registration Started • National Hackathon Registration Open • Semester Examination Timetable". The main content area is light gray and contains a white contact form. The form has a title "Contact Us" and a subtitle "We'd love to hear from you! Fill out the form below.". There are five input fields: "Full Name", "Email Address", "Phone Number", "Subject", and "Message". Below the message field are two buttons: "Send Message" (blue) and "Reset" (blue). At the bottom of the page, there is a link "Contact Information" and a location pin icon followed by the text "CHARUSAT University, Chanao, Anand, Gujarat".

Figure 3.7(Contact us page)

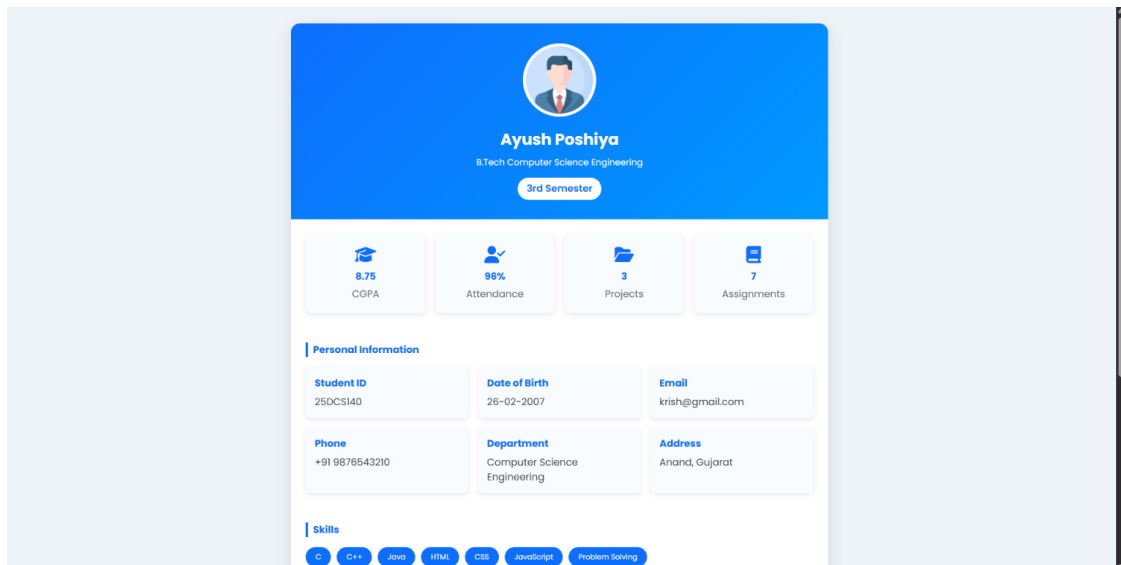


Figure 3.8(Profile page)

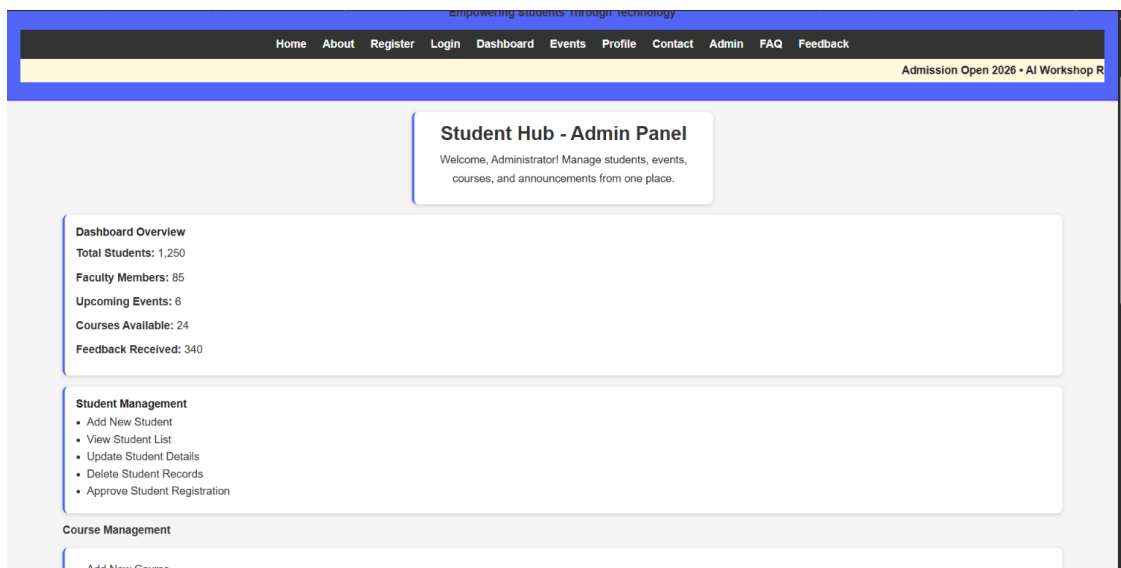


Figure 3.9(Admin page)

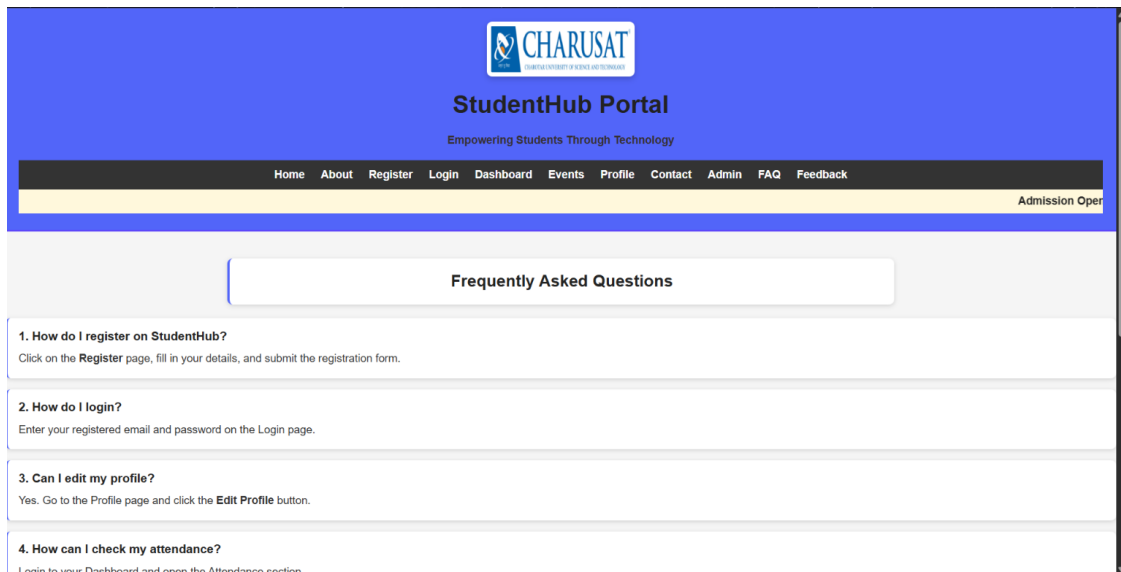


Figure 3.10(Faq page)

The screenshot shows the Student Hub Feedback Form page. At the top, there is a blue header with the text "Empowering Students Through Technology". Below this is a navigation bar with links: Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. A yellow banner below the navigation bar says "Admission Open 2026 • AI Workshop Registration Started • National Hackathon Registration Open • Semes". The main content area is titled "Student Hub Feedback Form" and contains the text "Thank you for using Student Hub. Your valuable feedback helps us improve our platform and provide a better learning experience." Below this is a form with two sections:

**Personal Information**

Full Name :

Email Address :

Student ID :

Department :

Semester :

**Feedback Details**

Figure 3.11(Feedback page)

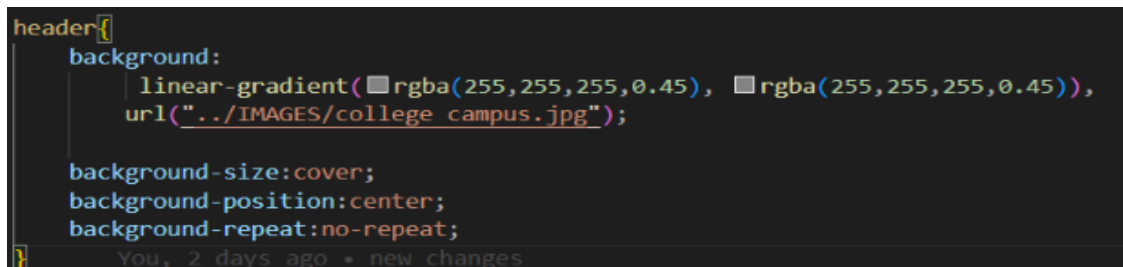
## I. GitHub Repository Link:

<https://github.com/ayushposhiya00-lab/Student-Hub>

**J. Challenges Faced:**

When I was styling the html files , I had no idea about how I put image behind the text and blur it . then I told ChatGPT that I want to put image behind text at header which concept or styling function I should use , give syntax and opacity code and gradient code like (255,255,255,0.20);

It gave me function that name is linear gradient and also I learned padding , card , form, Flex , grid, styling , box , responsive etc.. concept.

A screenshot of a code editor showing CSS code for a header. The code is as follows:

```
header{  
  background:  
    linear-gradient(rgba(255,255,255,0.45), rgba(255,255,255,0.45)),  
    url("../IMAGES/college campus.jpg");  
  background-size:cover;  
  background-position:center;  
  background-repeat:no-repeat;  
}
```

At the bottom of the editor, there is a status bar that says "You, 2 days ago • new changes".**K. Learning Outcomes:**

I will design a professional, responsive, and reusable UI layout.

**Faculty Remarks: (Keep this blank for faculty)**